

Art Exhibition Management System (AEMS)

Group 12 - Abrar Makki, Brian Daza, and Aditi Ghosh

Project Description

Art Exhibition Management System (AEMS) is a web-based application developed to modernize and streamline the management of art exhibitions. The system is designed to support both administrative users such as gallery managers and organizers, and customers such as visitors and art buyers. Its primary goal is to provide a centralized platform for managing artworks, exhibitions, ticket sales, and auctions while enhancing user interaction with art content.

Admin and Customer Users

AEMS supports two main user roles: customers and admins. Customers can create accounts, browse artworks, view exhibitions, purchase tickets, and participate in auctions. The system also allows users to explore artworks in detail, including support for 3D model visualization. Admin users have access to a separate dashboard that enables them to manage artworks, exhibitions, auctions, customer data, and reports. This role-based access control ensures that system operations remain secure and organized.

Techniques, Methodologies, and Tools

- **Pair programming:** Used such that two team members worked together at one laptop, with one writing code and the other reviewing to improve quality and reduce errors.
- **Small releases:** Fixes and features added in small increments and updates to the project.
- **Collective code:** Each member had equal access to the code, ensuring that each member had shared collaboration and responsibility across the project.
- **Weekly Scrum meetings:** Short weekly meetings where the team discussed progress, upcoming tasks, and issues to be addressed.
- **Jira (Kanban):** Used to organise tasks on a visual board, track progress through different stages (To Do, In Progress, Done), and manage progress efficiently.
- **VSCode:** The main code IDE and editor used for development, using backend extensions and integrated Git control.
- **GitHub:** Used for version control and collaboration, allowing the team to store code, manage changes, and work on different tasks before merging updates.

Development Process

The project was delivered in three phases:

Release 1: Established the customer side of AEMS, including the welcome page, registration, login, customer dashboard, artwork browsing, upcoming events, ticket purchasing, cart, and payment workflow. This release focused on allowing customers to explore exhibitions and artwork and complete basic purchase-related actions. Some planned features, such as automatic ticket delivery by email or phone, were not completed.

Release 2: Expanded the system to include the admin side and more advanced features. This release added the admin dashboard, artwork management, exhibition management, order management, customer management, auction management, bidding functionality, 3D artwork viewing, and reports/analytics. Some advanced features, such as real-time auction updates, automatic auction winner notification, and full report exporting, were not completed.

Release 3: Refined the existing customer and admin pipelines and expanded the reporting and analytics features. This release focused on improving the admin dashboard, reports page, transaction/order visibility, exhibition displays, customer profile history, and 3D artwork experience. Admin users could view statistics related to orders, tickets, auctions, customers, revenue, top artworks, and upcoming exhibitions. Customers could continue browsing artworks and events, view more detailed exhibition information, purchase tickets or artwork, and see purchase history through their profile. This release helped connect customer activity more clearly to admin reports and system records. Some planned refinements, such as full real-time analytics, support ticket handling, and more advanced multi-artist exhibition management, were not fully completed and remain for future improvements.

Future Work

Future improvements for AEMS include integrating a real payment system, implementing email and SMS notifications, enhancing security measures, and supporting real-time auction updates for users. Expanding the 3D viewing experience into a virtual gallery environment and adding personalized recommendations are also potential enhancements that could significantly improve user engagement.

Conclusion

In conclusion, the Art Exhibition Management System successfully demonstrates the core functionalities required for managing digital art exhibitions. While the current scope of the project provides a viable prototype, further development and refinements are necessary to transform it into a fully production-ready system available to customers, artists, and exhibition organizers.